

Girijananda Chowdhury University, Guwahati

Department of Civil Engineering

B.Tech 1st Semester

End semester Examination

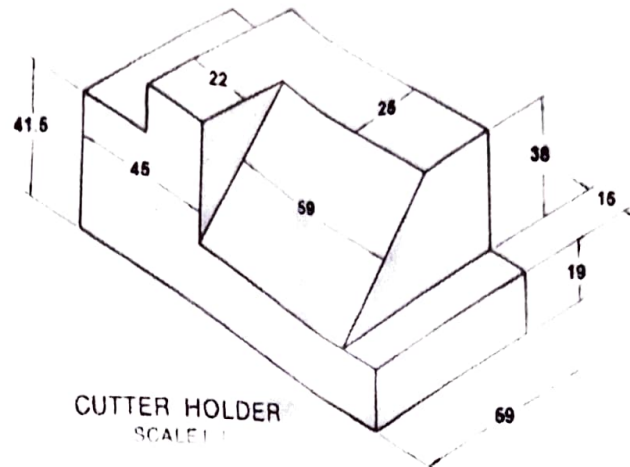
Paper Code: BCE23101P

Subject- **Engineering Graphics and Design**

Time: 2 Hrs and 30 Mins

Total Marks: 60

1. Draw the following in Single Stroke Capital inclined letters of 35 mm size "GIRIJANANDA CHOWDHURY UNIVERSITY" 10
2. Draw rectangular or equilateral hyperbola given the position of point P on it, 40mm and 20mm from the asymptotes OX and OY respectively. 10
3. A) Construct a hyperbola when the distance between the focus and directrix is 50 mm. Draw the tangent and normal at any point on the hyperbola 5
B) Draw projection of a point N, 40 mm below the HP and on the VP 5
4. A Pentagonal Prism side of base 30mm and axis 50mm long stands on HP on its base with one of the side of base perpendicular to VP. It is cut by a sectional plane 45 degree incline to HP and passing through the centre of the axis. Draw the front view, sectional top view and the sectional side view. Also draw the true shape of section and the development of the surface of the remaining solid. 10
5. Construct a hyperbola when the distance between the focus and directrix is 50 mm. Draw a tangent and normal at any point on the hyperbola. 10
6. Draw an ellipse of major axis 100mm and minor axis 50mm by arc of circle method. Also draw the tangent and normal at any point on the curve. 10
7. The one side of a regular hexagon of 35 mm side is in HP while the opposite side is 55 mm above the HP and incline 40 degree to VP. Draw the projection of plane and find surface inclination with HP. 10
8. Figure shows the pictorial view of an object, draw the following views 10
 - i. Front view looking from the direction of X.
 - ii. Top view.
 - iii. Side view



9. Construct a diagonal scale of 1:50 to read metre, decimetre and centimetre and long enough to measure up to 5 metre. Mark on it a distance of 4.94 metre.